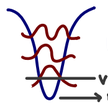
Hydrogen Atom

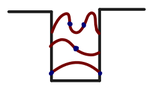


$U(r) \sim \frac{-2e^2}{r}$ remember: $\left(\frac{1}{r}\right)$

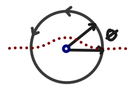
$E \propto -\frac{1}{N^2}$ "principle quantum number" $-n$ (first quantum number)

extra protons → + core electrons

finite barrier



electronic orbits



$\psi(r, \theta, \phi)$

↳ $N\lambda = 2\pi r \rightarrow \lambda = \frac{2\pi r}{N}$

• $k = \frac{2\pi}{\lambda} = \frac{2\pi}{2\pi r} N$

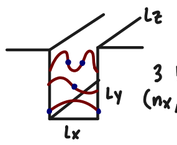
⇒ $L = rp = \frac{N}{p} r = N\hbar$, $p = \hbar k$

"angular momentum quantum number l" → states combined can lead to additional quantum number

3D orbits

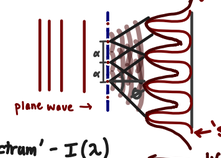
orientation - direction that orbit points + (quantum number m)

spin (4D number) - 'orientation + time' (quantum number s)



3 DOF (n_x, n_y, n_z)

Spectroscopy

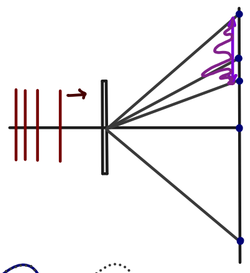


plane wave →

'spectrum' - $I(\lambda)$

← 'sharp lines' - correspond to eigenstates of electron

$I(\theta)$ diffraction grating (non quantum effect)

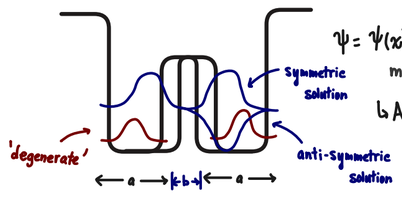


2nd order

1st order

$\sin\theta = \frac{N\lambda}{a}$

Quantized energy states correspond to energy level eigenstates



'degenerate'

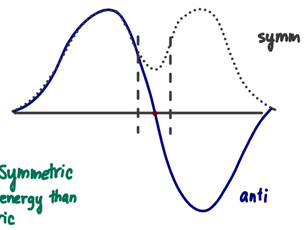
Symmetric solution

anti-symmetric solution

$\psi = \psi(x) e^{i(\omega t + \theta)}$

middle region

↳ $Ae^{-kx} + Be^{kx}$



symm

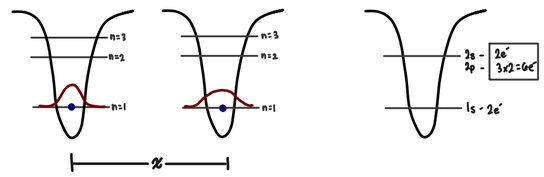
anti

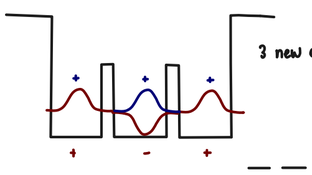
* In general, Symmetric solution has lower energy than antisymmetric

Symmetric has slightly longer λ

→ $k = \frac{2\pi}{\lambda}$ $E = \frac{\hbar^2 k^2}{2m}$

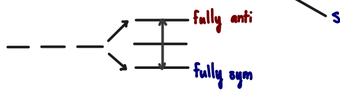
atomic model





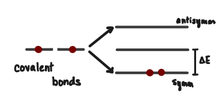
3 new eigenstates:

$E_1 E_2$ → antisymmetric, symmetric



fully anti

fully sym



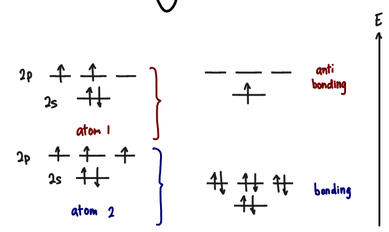
covalent bonds

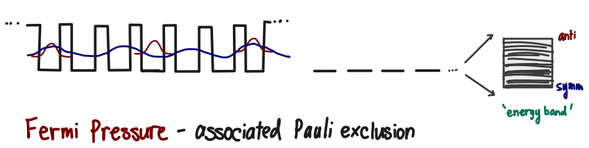
antisymm

symm

ΔE

* $\frac{\Delta E}{\Delta x}$ → force

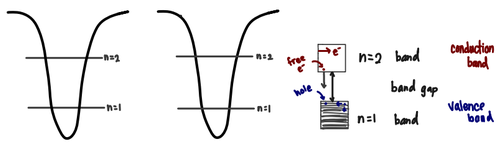




anti

symm

'energy band'



Partially filled band

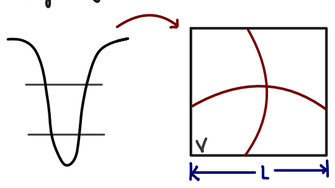
pauli-exclusion - "fermi-sea"

: metal/conductor - no gap for topmost e⁻

Fermi Pressure - associated Pauli exclusion energy

why doesn't the electron energy fall to zero?

'why things feel solid'

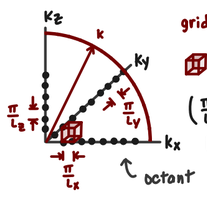


approximate states as standing waves in a fixed rectangular box

① count total # of 'states' in volume V.

L - direction: $\lambda = \frac{2L}{N} \Rightarrow k_x = \frac{2\pi}{\lambda} = \frac{2\pi}{2L_x} N_x = \frac{\pi}{L_x} N_x$

3D volume: $k_y = \frac{\pi}{L_y} N_y$, $k_z = \frac{\pi}{L_z} N_z$



grid: $\left(\frac{2\pi}{L_x} N_x, \frac{2\pi}{L_y} N_y, \frac{2\pi}{L_z} N_z\right)$

▢ - volume in k-space

$\left(\frac{\pi}{L_x}\right)\left(\frac{\pi}{L_y}\right)\left(\frac{\pi}{L_z}\right) = \frac{\pi^3}{V}$

↳ volume per state in k-space

② count total number of fermions in the least states ⇒ = $\frac{\text{total volume of k-sphere}}{\text{volume per state in k-sphere}}$

$E_F = \frac{\hbar^2 k_F^2}{2m} \Rightarrow N = \left(\frac{V}{\pi^2}\right) 2 \cdot \frac{1}{8} \int_0^{k_F} 4\pi k^2 dk$

total # of electrons; 2 spin state; one octet of 3D space

$= \left(\frac{V}{\pi^3}\right) \frac{1}{4} \int_0^{E_F} 4\pi \frac{k^2 m}{\hbar k} dE$

$= \left(\frac{V}{\pi^3}\right) \frac{1}{4} \int_0^{E_F} 4\pi \frac{\sqrt{2mE}}{\hbar^2} m\, dE$

$E = \frac{\hbar^2 k^2}{2m}$

$\frac{dE}{dk} = \frac{\hbar^2 k}{m}$

$dE = \frac{\hbar k}{m} dk$

$dk = \frac{m}{\hbar^2 k} dE$

$k = \frac{\sqrt{2mE}}{\hbar}$

density of particles: $\frac{N}{V} = \frac{2\sqrt{2}}{2\pi^2} \frac{m^{3/2}}{\hbar^3} \left(\frac{2}{3} E_F^{3/2}\right)$ = n

③ equate num particles to natural energy

↳ take inverse: $E_F = \left(n \frac{(3\pi^2)}{2\sqrt{2}} \frac{\hbar^3}{m^{3/2}}\right)^{2/3}$

$= \left(n \frac{(3\pi^2)(\hbar c)^3}{2\sqrt{2}(mc^2)^{3/2}}\right)^{2/3}$

$\sim 10^{14}$ engineer's method

$\hbar c = 2 \times 10^{-5}$ eV·cm (E/cm)

$mc^2 = 5 \cdot 10^5$ eV (for e⁻)

'typical' density of atoms $= \sim \frac{10^{23}}{cm^3}$

↳ $\left[\frac{\frac{10^{23}}{cm^3} \cdot 10 (2 \cdot 10^{-5} eV \cdot cm)^3}{(5 \cdot 10^6 eV)^{3/2}}\right] = \left(\frac{10^{23} \cdot 10^2 \cdot 10^{-15}}{10^9}\right)^{2/3} \simeq 10^{2/3}$ eV ≃ 1-10 eV!

Pauli Exclusion Energy:

Suppose we compress V by 1% - how much pressure does that require?

$V = \int_0^{E_F} \frac{V}{\pi^2} \frac{\sqrt{2} \cdot m^{3/2}}{\hbar^2} E^{1/2} \cdot E\, dE$

↑ weight the number integral by the energy per particle

$= 2V \frac{\sqrt{2}}{5\pi^2} \frac{m^{3/2}}{\hbar^2} \left(\frac{N}{V} \frac{(3\pi^2)}{2\sqrt{2}} \frac{\hbar^3}{m^{3/2}}\right)^{5/3} \propto \frac{1}{V^{2/3}}$

* $P = -\frac{dU}{dV} \Rightarrow \frac{\text{total change in energy}}{\text{total change in volume}}$

↳ $P = \frac{(\text{Force} - \text{length})}{\text{Area} - \text{length}}$, $P = \frac{2}{3} \frac{U}{V}$

"compressibility" $B = -V \frac{dP}{dV} = \frac{5}{3} P = \frac{2}{3} n E_F$

'strain'

↳ $\Delta P = B\left(\frac{\Delta V}{V}\right) = \frac{2}{3} n E_F (.01)$

$= \frac{2}{3}\left(\frac{10^{23}}{cm^3} 10 eV\right)(.01) = \frac{2}{3}\left(\frac{10^{23}}{cm^2} \cdot 10 eV \frac{J}{6 \cdot 10^8 eV}\right)(.01)$

10^5; $10^3 \frac{J}{cm^3}$

• $J = 624 \times 10^{18}$ eV * $J = N \cdot m$

↳ $N = \frac{J}{m} \frac{m}{100cm} = .01 \frac{J}{cm}$

$1 atm \simeq 10 \frac{N}{cm^3} = \frac{10 \times .01}{cm^3} J = .01 \frac{J}{cm^3}$

	Spin	E - Eigenstates	particles	rate of
Fermions	$\pm\frac{1}{2}, \pm\frac{3}{2}, \ldots$	$E = \frac{1}{2}\hbar\omega N$ (N: 0 or 1)	e^-, p^+, n, q	$\propto (1 + N_f)$ 'pauli exclusion'
Bosons	$0, \pm 1, \pm 2, \ldots$	$E = \hbar\omega N$	γ, phonons	rate of $i \rightarrow f$ $\propto (1 + N_f)$

'stimulated' transitions

'spontaneous' transitions

Summary of solid QM

① Boundary of atoms in molecules occurs a. QM effect of symm & antisymm superpositions

b. charge transfer - ionic bond

② solids are built from molecules in same ways

→ atomic orbitals overlap

→ "bands" and 'band gaps'

③ bands correspond to 'extended states'

→ shared by many atoms

• classically electrons are thought of as jumping between orbital shells, in QM - the movements are modeled as one uniform traveling wave in this extended state ≃ a 'bumpy' plane wave

④ Pauli exclusion plays a major rule - electrons fill these states from lowest energy up ∼ 'Fermi Sea' ala 'Dirac Sea'

⑤ Semi Conductors have a gap between highest filled energy + lowest unfilled, metals do not

⑥ Free electrons in conduction band + holes in volume

• both carrying charge in a current

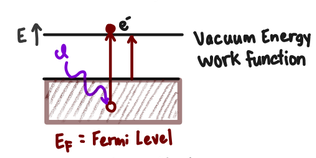


photo-electric effect

2 experiments that led to QM that has 'Nothing to do w/ Particles'

① photoelectric Effect

② Planck's Radiation Spectrum

} breakdown of 'classical' wave particle

Results from experiments on photo electric effect

① Threshold Wavelength - minimum photon energy (ψ_f - wave frequency)

② current 'turns on' instantaneously - regardless of light intensity

③ current proportional to light intensity

→ modern revelations / Updates to Understanding

• There is actually a 'turn-on' time: ∼ picosecond - still independent of intensity

τ ← 'decoherence time' (phase scrambling time)

• Theory of Bands: electrons in wave states across whole crystal